2. the *parallel transport* $\prod_c^\nabla$ which defines a way to transport vectors between tangent planes along any smooth curve c,

3. the notion of ∇-geodesics γ_∇ which are defined as autoparallel curves, thus extending the ordinary notion of Euclidean straightness,

4. the intrinsic curvature and torsion of the manifold.

2.2 Metric tensor fields g

The *tangent bundle* of M is defined as the "union" of all tangent spaces:

$$TM := \cup_p T_p = \{(p, v), \quad p \in M, v \in T_p\}. \tag{1}$$

Thus the tangent bundle TM of a D-dimensional manifold M is of dimension $2D$. (The tangent bundle is a particular example of a fiber bundle with base manifold M.)

Informally speaking, a *tangent vector* v plays the role of a directional derivative, with vf informally meaning the derivative of a smooth function f (belonging to the space of smooth functions $\mathfrak{F}(M)$) along the direction v. Since the manifolds are abstract and not embedded in some Euclidean space, we do not view a vector as an "arrow" anchored on the manifold. Rather, vectors can be understood in several ways in differential geometry like directional derivatives or equivalent class of smooth curves at a point. That is, tangent spaces shall be considered as the manifold abstract too.

A smooth *vector field* X is defined as a "cross-section" of the tangent bundle: $X \in \mathfrak{X}(M) = \Gamma(TM)$, where $\mathfrak{X}(M)$ or $\Gamma(TM)$ denote the space of smooth vector fields. A basis $B = \{b_1, \ldots, b_D\}$ of a finite D-dimensional vector space is a *maximal linearly independent set of vectors*: A set of vectors $B = \{b_1, \ldots, b_D\}$ is linearly independent if and only if $\sum_{i=1}^D \lambda_i b_i = 0$ iff $\lambda_i = 0$ for all $i \in [D]$. That is, in a linearly independent vector set, no vector of the set can be represented as a linear combination of the remaining vectors. A vector set is linearly independent maximal when we cannot add another linearly independent vector. Tangent spaces carry algebraic structures of vector spaces. Furthermore, to any vector space V, we can associate a dual covector space V^* which is the vector space of real-valued linear mappings. We do not enter into details here to preserve this gentle introduction to information geometry with as little intricacy as possible. Using local coordinates on a chart $(\mathcal{U}, x)$, the vector field X can be expressed as $X = \sum_{i=1}^D X^i e_i \overset{\Sigma}{=} X^i e_i$ using Einstein summation convention on dummy indices (using notation $\overset{\Sigma}{=}$), where $(X)_B := (X^i)$ denotes the *contravariant vector components* (manipulated as "column vectors" in algebra) in the *natural basis* $B = \{e_1 = \partial_1, \ldots, e_D = \partial_D\}$ with $\partial_i :=: \frac{\partial}{\partial x_i}$. A tangent plane (vector space) equipped with an *inner product* $\langle \cdot, \cdot \rangle$ yields an *inner product space*. We define a *reciprocal basis* $B^* = \{e^{*i} = \partial^i\}_i$ of $B = \{e_i = \partial_i\}_i$ so that vectors can also be expressed using the *covariant vector components* in the natural reciprocal basis. The primal and reciprocal basis are *mutually orthogonal* by construction as illustrated in Figure 2.

For any vector v, its contravariant components v^i's (superscript notation) and its covariant components v_i's (subscript notation) can be retrieved from v using the inner product with the use of the reciprocal and primal basis, respectively:

$$v^i = \langle v, e^{*i} \rangle, \tag{2}$$

$$v_i = \langle v, e_i \rangle. \tag{3}$$

The inner product defines a *metric tensor* g and a *dual metric tensor* g^*:

$$g_{ij} := \langle e_i, e_j \rangle, \tag{4}$$

$$g^{*ij} := \langle e^{*i}, e^{*j} \rangle. \tag{5}$$

Technically speaking, the metric tensor $g_p : T_pM \times T_pM \to \mathbb{R}$ is a 2-covariant tensor field:

$$g \overset{\Sigma}{=} g_{ij} \mathrm{d}x_i \otimes \mathrm{d}x_j, \tag{6}$$

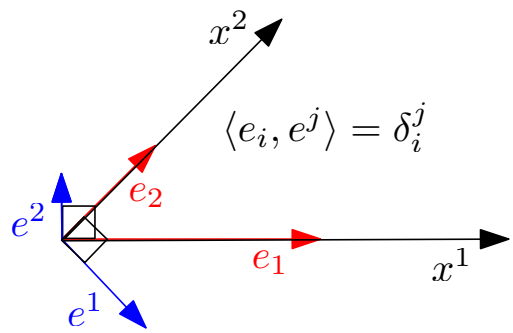

Figure 2: Primal basis (red) and reciprocal basis (blue) of an inner product $\langle \cdot, \cdot \rangle$ space. The primal/reciprocal basis are mutually orthogonal: e^1 is orthogonal to e_2, and e_1 is orthogonal to e^2.

where $\otimes$ is the dyadic tensor product performed on pairwise covector basis $\{\mathrm{d}x_i\}_i$ (the covectors corresponding to the reciprocal vector basis). We do not describe tensors in details for sake of brevity. A tensor is a geometric entity of a tensor space that can also be interpreted as a multilinear map. A contravariant vector lives in a vector space while a covariant vector lives in the dual covector space. We recommend the textbook [66] for a concise and well-explained description of tensors.

Let $G = [g_{ij}]$ and $G^* = [g^{*ij}]$ denote the $D \times D$ matrices It follows by construction of the reciprocal basis that $G^* = G^{-1}$. The reciprocal basis vectors e^{*i}'s and primal basis vectors e_i's can be expressed using the dual metric g^* and metric g on the primal basis vectors e_j's and reciprocal basis vectors e^{*j}'s, respectively:

$$e^{*i} \overset{\Sigma}{=} g^{*ij} e_j, \tag{7}$$

$$e_i \overset{\Sigma}{=} g_{ij} e^{*j}. \tag{8}$$

The *metric tensor field* g ("metric tensor" or "metric" for short) defines a smooth symmetric positive-definite *bilinear form* on the tangent bundle so that for $u, v \in T_p, \quad g(u, v) \geq 0 \in \mathbb{R}$. We can also write equivalently $g_p(u, v) := \langle u, v \rangle_p := \langle u, v \rangle_{g(p)} := \langle u, v \rangle$. Two vectors u and v are said orthogonal, denoted by $u \perp v$, iff $\langle u, v \rangle = 0$. The length of a vector is induced from the *norm* $\|u\|_p := \|u\|_{g(p)} = \sqrt{\langle u, u \rangle_{g(p)}}$. Using local coordinates of a chart $(\mathcal{U}, x)$, we get the vector contravariant/covariant components, and compute the metric tensor using matrix algebra (with column vectors by convention) as follows:

$$g(u, v) = (u)_B^\top \times G_{x(p)} \times (v)_B = (u)_{B^*}^\top \times G_{x(p)}^{-1} \times (v)_{B^*}, \tag{9}$$

since it follows from the primal/reciprocal basis that $G \times G^* = I$, the identity matrix. Thus on any tangent plane T_p, we get a *Mahalanobis distance*:

$$M_G(u, v) := \|u - v\|_G = \sqrt{\sum_{i=1}^{D} \sum_{j=1}^{D} G_{ij} (u^i - v^i)(u^j - v^j)}. \tag{10}$$

The inner product of two vectors u and v is a scalar (a 0-tensor) that can be equivalently calculated as:

$$\langle u, v \rangle := g(u, v) \overset{\Sigma}{=} u^i v_i \overset{\Sigma}{=} u_i v^i. \tag{11}$$

A metric tensor g of manifold M is said *conformal* when $\langle \cdot, \cdot \rangle_p = \kappa(p) \langle \cdot, \cdot \rangle_{\text{Euclidean}}$. That is, when the inner product is a scalar function $\kappa(\cdot)$ of the Euclidean dot product. More precisely, we define the notion of a metric g' conformal to another metric g when these metrics define the same angles between vectors u and v of a tangent plane T_p:

$$\frac{g'_p(u, v)}{\sqrt{g'_p(u, u)} \sqrt{g'_p(v, v)}} = \frac{g_p(u, v)}{\sqrt{g_p(u, u)} \sqrt{g_p(v, v)}}. \tag{12}$$

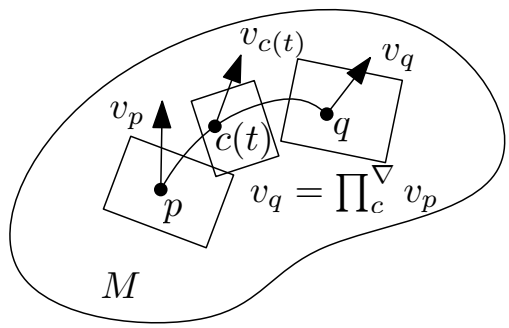

Figure 3: Illustration of the parallel transport of vectors on tangent planes along a smooth curve. For a smooth curve c, with $c(0) = p$ and $c(1) = q$, a vector $v_p \in T_p$ is parallel transported smoothly to a vector $v_q \in T_q$ such that for any $t \in [0, 1]$, we have $v_{c(t)} \in T_{c(t)}$.

Usually g' is chosen as the Euclidean metric. In conformal geometry, we can measure angles between vectors in tangent planes as if we were in an Euclidean space, without any deformation. This is handy for checking orthogonality in charts. For example, Poincaré disk model of hyperbolic geometry is conformal but Klein disk model is not conformal (except at the origin), see [89].

2.3 Affine connections ∇

An affine connection ∇ is a differential operator defined on a manifold that allows us to define (1) a covariant derivative of vector fields, (2) a parallel transport of vectors on tangent planes along a smooth curve, and (3) geodesics. Furthermore, an affine connection fully characterizes the curvature and torsion of a manifold.

2.3.1 Covariant derivatives $\nabla_X Y$ of vector fields

A connection defines a *covariant derivative* operator that tells us how to differentiate a vector field Y according to another vector field X. The covariant derivative operator is denoted using the traditional gradient symbol ∇. Thus a covariate derivative ∇ is a function:

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M), \tag{13}$$

that has its own special subscript notation $\nabla_X Y := \nabla(X, Y)$ for indicating that it is differentiating a vector field Y according to another vector field X.

By prescribing D^3 smooth <u>functions</u> $\Gamma_{ij}^k = \Gamma_{ij}^k(p)$, called the *Christoffel symbols of the second kind*, we define the unique *affine connection* ∇ that satisfies in local coordinates of chart $(\mathcal{U}, x)$ the following equations:

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k. \tag{14}$$

The Christoffel symbols can also be written as $\Gamma_{ij}^k := (\nabla_{\partial_i} \partial_j)^k$, where $(\cdot)^k$ denote the k-th coordinate. The k-th component $(\nabla_X Y)^k$ of the covariant derivative of vector field Y with respect to vector field X is given by:

$$(\nabla_X Y)^k \overset{\Sigma}{=} X^i (\nabla_i Y)^k \overset{\Sigma}{=} X^i \left(\frac{\partial Y^k}{\partial x^i} + \Gamma_{ij}^k Y^j \right). \tag{15}$$

The Christoffel symbols are *not* tensors (fields) because the transformation rules induced by a change of basis do not obey the tensor contravariant/covariant rules.